

Reflective Journal

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In this lab, we explored how ensemble learning can improve a machine learning model by comparing a Decision Tree Classifier with a Random Forest Classifier using the Iris dataset. We trained both models in Python using scikit-learn, evaluated their performance, and created a feature importance plot to better understand which features had the greatest impact on the model's prediction. One of the biggest takeaways from this lab was seeing how a Random Forest combines the prediction from multiple decision trees to reduce overfitting and create a more reliable model. We also learned how feature importance helps us identify which variables contribute the most to a model instead of simply accepting the predictions without understanding what drives them.

One thing that surprised us was that both the Decision Tree and the Random Forest performed better. This shows us that a more advanced algorithm does not always produce better results, especially when working with a small, clean dataset like the Iris dataset. We also found that petal width was the most important feature, followed by petal length, while sepal measurement had much less influence on the model's predictions. Overall, this lab helped us better understand when ensemble methods are useful, and moving forward we would like to apply these techniques to larger, more complex datasets where the advantages of Random Forest would likely become noticeable.